

HardStance at StanceEval-2026: Error-Driven Prompt Calibration for Arabic Stance Detection

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Abstract

We present our system for StanceEval-2026 Track 1, which focuses on Arabic stance detection toward *Women Driving* using the Mawqif-XT dataset with three labels: *Favor*, *Against*, and *None*. We compare fine-tuned AraBERT models with zero-shot Gemma-4-31B. We refine the Gemma prompt using recurring development-set errors, focusing on dialect and indirect meaning, target identification, sentiment versus stance, hashtags, sarcasm, and the distinction between reporting and endorsement. We also compare the calibrated system with several alternative inference strategies, including self-consistency, model debate, and dialect-interpreter pipelines. The final Gemma-4-31B system achieves 86.61% Fav2, 67.46% Fav3, and 82.67% accuracy on the official blind test, ranking 5th. Our results show that target-specific context and clear decision rules can be more effective than adding additional inference stages or model interactions. The main remaining challenge is distinguishing *Against* from *None*, particularly when stance is indirect, ambiguous, or expressed through humor.

1 Introduction

StanceEval-2026 is a shared task on Arabic stance detection. The goal is to classify a writer’s stance toward a given target as *Favor*, *Against*, or *None* (Albalawi et al., 2026a). This task is challenging for Arabic social media. Users often write in different dialects and express their opinions indirectly. Sarcasm, rhetorical expressions, and hashtags can also change the intended meaning. In addition, cultural and historical context may be needed to understand a stance. A negative statement does not always indicate opposition to the target. For example, a user may criticize how a policy is implemented while still supporting the policy itself.

We participate in Track 1 using the Mawqif-XT dataset (Albalawi et al., 2026b)¹. The training and

development data cover COVID-19 Vaccine, Digital Transformation, and Women Empowerment. Women Driving is the evaluation target. We compare supervised AraBERT models with zero-shot generative LLMs, mainly Gemma and Qwen. We then refine the LLM prompts based on common errors observed during development. The refinement focuses on dialect and indirect meaning, target identification, sentiment versus stance, sarcasm, hashtags, and the difference between reporting an event and supporting it. We also test several alternative inference strategies, including self-consistency, translation-based classification, dialect interpretation, and multi-model debate.

The main contributions of this work are as follows:

- We compare fine-tuned AraBERT models with zero-shot LLMs for Arabic target-specific stance detection.
- We develop a prompt-calibration strategy based on common stance detection errors, including dialect, indirect stance, target confusion, sarcasm, hashtags, and reporting versus endorsement.
- We compare the calibrated single-model approach with several multi-stage and multi-model inference strategies.
- Our final Gemma-4-31B system achieves 86.61% Fav2, 67.46% Fav3, and 82.67% accuracy on the official Women Driving test set, ranking 5th in the shared task.

2 Background

Arabic stance detection has mainly been studied using supervised Transformer-based models. AraBERT (Antoun et al., 2020) is widely used for Arabic language understanding, while AraStance (Alhindi et al., 2021) provides a benchmark for Arabic stance detection across different countries and <https://github.com/abdulehsan/StanceEval>.

¹Code and experimental resources are available at

domains. Previous surveys also summarize common methods, datasets, and challenges in stance detection (ALDayel and Magdy, 2021; Albalawi et al., 2026b).

More recent studies have explored large language models for stance classification. One direction is to provide target-specific background knowledge to improve zero-shot prediction (Zhu et al., 2022). Other work studies prompting and fine-tuning open-source LLMs for stance classification (Cruickshank and Ng, 2026). These studies show that LLM performance can depend strongly on how the task and target are presented in the prompt. In addition, rephrasing the input before answering has been shown to improve LLM performance on some tasks (Deng et al., 2024). This motivates our use of a mental-rewrite step to help the model interpret dialectal, sarcastic, and rhetorical expressions.

Building on this work, we compare supervised AraBERT models with generative LLMs for Arabic social-media stance detection. Our main focus is **systematic prompt calibration**. We refine the prompt to address common sources of ambiguity, including target identification, dialect interpretation, target-specific historical context, sarcasm, hashtag use, and the distinction between reporting an event and expressing support for it.

3 System Overview

We investigate two paradigms: supervised encoder fine-tuning and zero-shot generative LLM classification. We used encoder based AraBERT as baseline and soon moved to the zero-shot LLM classification upon getting better results.

3.1 Supervised Baseline: AraBERT

We fine-tune three AraBERT variants (base, Twitter-pretrained base, Twitter-pretrained large) (Antoun et al., 2020), each receiving the target and tweet as paired input and predicting *Favor*, *Against*, or *None*. Class-weighted cross-entropy compensates for substantial class imbalance, particularly the underrepresented *None* class. These models establish a baseline for comparing task-specific fine-tuning against prompting a larger generative model.

3.2 Zero-Shot LLM Classification

We use Gemma-4-31B (Team, 2026), Qwen3.6-27B (Qwen Team, 2026) as zero-shot stance classifiers. While GPT-5.6-sol and GPT-5.6-luna were

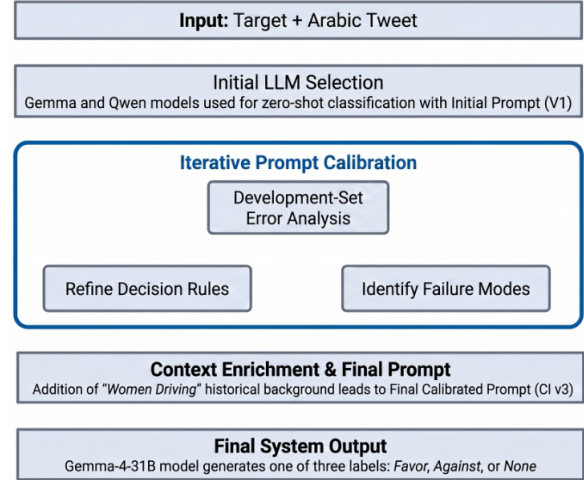


Figure 1: Iterative prompt calibration using development-set errors to refine stance decision rules.

used with Context Injection. The models receive the target and tweet and predict one of three labels: *Favor*, *Against*, or *None*. We iteratively refine the prompt using errors observed on the development set, see figure 1. The refinements focus on recurring sources of ambiguity, including dialect and implicit meaning, target identification, sentiment versus stance, sarcasm and hashtag interpretation, and the distinction between reporting an event and expressing support for it. We also include a mental-rewrite step to help interpret dialectal and rhetorical expressions, following the rephrase-before-respond principle (Deng et al., 2024). The complete prompt and decision rules are provided in Appendix A.2.

3.2.1 Context Injection (CI) in Zero Shot Stance Classification

For the evaluation target, Women Driving, no target-specific labeled examples exist in Mawqif-XT’s training or development data. We inject concise historical context on the 2017–2018 Saudi driving-policy change, drawing on evidence that targeted background knowledge improves zero-shot stance generalization (Zhu et al., 2022), paired with the same stance framework and mental-translation procedure. Decision rules are calibrated using patterns observed during development phase.

3.3 Additional Systems

We compare the calibrated single-model system with several more complex inference strategies:

- **Self-Consistency (SC).** We use temperature-based sampling with early-stopping majority

Split	Targets	Size	Use
Train	Covid Vaccine,	3,502	AraBERT training
	Women Empowerment, Digital Transformation		
Dev	Same three targets	619	Prompt calibration
Test	Women Driving	352	Blind evaluation

Table 1: Mawqif-XT data used in StanceEval-2026.

voting to test whether multiple generations improve prediction consistency of Gemma-4-31B(Wang et al., 2023).

- **ALLaM Dialect Interpreter (ALLaM-DI).** We use a two-stage pipeline in which a locally hosted ALLaM-7B (Bari et al., 2025) model first interprets the Arabic dialect and context. The interpreted text is then passed to the stance classifier.
- **Explain-then-Classification (E-C).** The model first generates a brief explanation of the tweet’s intended meaning and stance-related cues, and then uses this interpretation to assign the final stance label. This tests whether an explicit reasoning step before classification improves stance prediction.
- **Gemma–Qwen–Gemma Debate (GQG-Debate).** Gemma produces an initial prediction, Qwen reviews it, and Gemma makes the final decision. This setup tests whether model interaction and additional reasoning improve classification (Du et al., 2024).

We compare our CI-based zero-shot classifier with these systems to examine whether additional reasoning stages, model diversity, or intermediate interpretation improve performance over the calibrated single-model approach. Comparative results are presented in Section 5, while full prompts and pipeline details are provided in the Appendix.

4 Experimental Setup

4.1 Dataset

We use the official Mawqif-XT dataset (Albalawi et al., 2026b) provided for StanceEval-2026. Table 1 summarizes the data splits and their use in our experiments.

4.2 Preprocessing

AraBERT inputs are processed with the official ArabertPreprocessor, with target and tweet given as a paired sequence. For LLM-based systems we deliberately avoid aggressive preprocessing, since normalization can remove dialectal, orthographic, hashtag, and emoji cues relevant to stance; the original tweet is preserved and any interpretation happens within the prompt itself.

4.3 Model and Inference Setup

We evaluate three AraBERT variants as supervised baselines and Gemma-4-31B and Qwen-3 variants as zero-shot stance classifiers. LLM inference uses deterministic decoding (temperature = 0.1, max_tokens = 10, no sampling). Calibrated configurations additionally include target-specific context and the decision rules described in Section 3.2. Gemma models are accessed through Cerebras Cloud, Qwen through Groq/OpenRouter, and ALLaM-7B is run locally for the dialect-interpretation pipeline. Full fine-tuning hyperparameters, hardware and software details, and pipeline configurations are provided in Appendix A.1.

5 Results and Discussion

5.1 Development-Set Results

Table 2 compares encoder and generative baselines on the full 619-example development set. Among fine-tuned encoders, AraBERTv0.2-Twitter-Large performs best but zero-shot Gemma-4-31B substantially outperforms all AraBERT variants, consistently across all three targets - motivating our focus on prompt calibration over further encoder fine-tuning.

5.2 Official Blind-Test Results

Table 3 reports the official blind-test results. The calibrated Gemma-4-31B CI system achieves the best performance across all three metrics, with 0.8661 Favg2, 82.67% accuracy, and 0.6746 Favg3. It improves substantially over the initial V1 prompt and also outperforms the more complex inference strategies. Among these alternatives, self-consistency performs best on Favg2, but remains below the calibrated single-model system. The dialect-interpreter and multi-agent debate pipelines perform considerably worse, suggesting that additional inference stages do not necessarily improve stance classification.

Model	COVID-19	Digital Transf.	Women Emp.	Favg2	Favg3
AraBERTv0.2-Base	0.7231	0.7160	0.8375	0.7855	0.6677
AraBERTv0.2-Twitter-Base	0.7808	0.7694	0.8627	0.8243	0.7129
AraBERTv0.2-Twitter-Large	0.8313	0.7282	0.8581	0.8346	0.7157
Gemma-4-31B	0.9066	0.7685	0.8996	0.8848	0.7820

Table 2: Development-set performance (619 examples), per-topic and overall.

System	Favg2	Accuracy	Favg3
<i>Prompt Calibration</i>			
Gemma-4-31B, V1	0.8063	75.28%	0.6183
Gemma-3.6-27B, V1	0.7758	73.01%	0.5862
Gemma-4-31B, V1 Revised	0.8444	80.11%	0.6429
<i>Alternative Inference Strategies</i>			
SC	0.8506	80.68%	0.6365
ALLaM-DI	0.7414	71.02%	0.5700
E-C	0.8296	77.84%	0.6034
GQG-Debate	0.8123	76.98%	0.5982
<i>LLM Comparisons with Context Injection (CI)</i>			
GPT-5.6-sol with CI	0.8308	77.56%	0.6529
GPT-5.6-luna with CI	0.8341	79.26%	0.6613
Gemma-4-31B with CI	0.8661	82.67%	0.6746

Table 3: Official results on the 352-example blind test set. The best score in each metric is shown in bold.

5.3 Discussion

Prompt development was driven by development-set error patterns. Four design decisions proved most important. Adding dialectal and historical context (Zhu et al., 2022) produced 26 prediction changes, predominantly *Against*→*Favor*, indicating the initial system over-relied on surface lexical cues. Requiring stronger evidence before assigning *Against* - while allowing *Favor/None* when opposition was only implied by jokes or ambiguous language - produced 25 further changes and reduced spurious opposition. An explicit reporting-versus-endorsement distinction reduced false *Favor* predictions on tweets merely describing policies or events. Finally, resolving the target before interpreting stance, and treating hashtags/sarcasm/rhetorical questions as contextual evidence rather than direct labels, addressed cases where a keyword suggested one position while the tweet’s intended meaning indicated another.

The strongest configuration combines historical context with calibrated rules for positive framing, target-directed criticism, and factual reporting. Full prompts are in Appendix A.2.

6 Error Analysis

Analysis of development-set errors and manual inspection of blind-test predictions reveal four recurring challenges. Because blind-test gold labels were withheld, the test-set cases discussed below and in Appendix A.4 are presented as illustrative

challenging examples identified through qualitative inspection, not confirmed misclassifications; only development-set examples have verified gold labels. **Hashtag hijacking:** hashtags can provide misleading stance cues, especially when they are used sarcastically or to criticize the opposing side. **Sarcasm and implicit stance:** stance expressed through mockery, rhetorical questions, or indirect language remains difficult to interpret, consistent with challenges reported for Arabic sarcasm detection (Abu Farha and Magdy, 2020). **Reporting versus endorsement:** factual mentions of a policy or event may be interpreted as support even when no clear stance is expressed. **Historical and cultural context:** some Women Driving tweets require knowledge of the 2017 Saudi driving-policy change and related social context to interpret implicit stance. A further challenge is the distinction between *Against* and *None*. Indirect criticism, ambiguous comments, or weak opposition cues can make it difficult to determine whether a tweet expresses a clear negative stance or no stance toward the target. These observations reinforce the distinction between sentiment and stance. Accurate classification requires identifying the target of the opinion and interpreting linguistic and contextual cues, particularly when the stance is indirect.

7 Conclusion

We compare fine-tuned AraBERT models with zero-shot generative LLMs for Arabic stance detection in StanceEval-2026. Our calibrated Gemma-4-31B system outperforms all AraBERT baselines and the alternative inference strategies tested in our experiments, ranking 5th on the official blind-test leaderboard. The results show the importance of clear contextual information and task-specific decision rules for interpreting sarcasm, hashtags, and implicit stance. More complex multi-stage and multi-model pipelines did not improve over the calibrated single-model approach. Distinguishing *Against* from *None* remains an important challenge, especially when the expressed stance is indirect or ambiguous.

Limitations

Error analysis (Section 6) is qualitative, since blind-test labels were withheld; dialect interpretation is constrained by limited Saudi dialect resources, and ALLaM-7B’s contribution as a local interpreter was not independently validated.

Ethical Considerations

This work uses the publicly available Mawqif-XT dataset for stance detection research; we did not collect new data or interact with human subjects. Tweets in the dataset may reflect deeply personal or politically sensitive opinions on a socially significant policy change; our system predicts stance for benchmarking purposes only and is not intended for deployment in content moderation, surveillance, or any application that could be used to profile or target individuals based on inferred political or social views. Stance detection systems, including ours, can misclassify sarcasm, dialectal expression, or minority viewpoints, and such errors could disproportionately affect speakers of underrepresented dialects if deployed without further validation.

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A Appendix

A.1 Experimental Configuration

AraBERT Fine-Tuning. We fine-tune three AraBERT variants: AraBERT-base, AraBERT-Twitter-base, and AraBERT-Twitter-large. Training uses AdamW with a learning rate of $2e-5$, batch size 16, weight decay 0.01, and a maximum of 5 epochs. We use early stopping with patience 2 and class-weighted cross-entropy to account for class imbalance.

LLM Inference. Gemma models are accessed through Cerebras Cloud, while Qwen models are accessed through Groq/OpenRouter. GPT models are accessed through Open AI API. ALLaM-7B is run locally using llama.cpp with the GGUF format for the dialect-interpretation pipeline. Unless otherwise stated, LLM inference uses deterministic decoding with temperature = 0.1, max_tokens = 10, and no sampling.

Hardware and Software. AraBERT-base experiments are run on an NVIDIA RTX 3050 Ti with 4GB VRAM, while AraBERT-Twitter-Large experiments use a Tesla T4 with 16GB VRAM. The software environment includes Python 3.11, PyTorch 2.0, Hugging Face Transformers 4.41, Pandas 2.0, and scikit-learn 1.2.

A.2 Prompt Specifications

We evaluated prompt configurations iteratively, using development-set errors to identify recurring failure modes and refine decision rules. To keep the main paper concise, we report the key configurations here.

System Prompt - V1

You are an expert annotator for Arabic stance detection. Given an Arabic tweet and a target topic, classify the writer's stance toward that target as exactly one of three labels:

- Favor: the writer expresses support for or a positive position toward the target
- Against: the writer expresses opposition to or a negative position toward the target
- None: no clear stance - neutral, off-topic, or ambiguous. This includes cases where sarcasm makes the literal tone misleading about the writer's actual position - do not infer a stance from tone alone if the underlying position isn't clear.

Respond with ONLY a single word: Favor, Against, or None. No explanation, no punctuation, no other text.

System Prompt - V1 Revised

You are an expert annotator for Arabic stance detection.

Task

Given an Arabic tweet and a target topic, classify the writer's stance toward the target as exactly one of:

- Favor
- Against
- None

Before assigning a stance, mentally rewrite the tweet into its intended literal meaning while preserving the writer's opinion, sarcasm, dialect, rhetorical intent, and emojis. Then determine the stance toward the target itself, not toward other people, quoted opinions, related entities, or hashtags.

Guidelines

- Favor: supports, defends, promotes, or welcomes the target.
- Against: opposes, criticizes, rejects, or mocks the target.
- None: no clear stance toward the target.

Important

- Determine where praise or criticism is directed. Negative language toward opponents of the target is usually Favor, not Against.
- Hashtags may be ironic or hijacked. Never infer stance from hashtags alone.
- Rhetorical questions, sarcasm, and emojis often convey the writer's true stance. Interpret the intended meaning rather than the literal wording.
- Distinguish reporting from endorsement. Mentioning an event or policy does not by itself express a stance.
- If the stance toward the target cannot reasonably be inferred, output None.

Respond with ONLY one word: Favor, Against, or None. Do not provide any explanation, punctuation, or additional text.

System Prompt - CI

You are an expert annotator for Arabic stance detection.

Background Context

The target concerns the 2017–2018 Saudi policy change allowing women to drive. Before June

2018, women were prohibited from driving in Saudi Arabia. Tweets from this period often discuss the royal decree, implementation, licensing, religion, tradition, safety, gender roles, media coverage, and public reaction. They frequently use Saudi dialect, sarcasm, humor, rhetorical questions, and indirect expressions.

Task

Given an Arabic tweet and a target topic, classify the writer's stance toward the target as exactly one of:

- Favor
- Against
- None

Before assigning a stance, mentally rewrite the tweet into its intended literal meaning while preserving the writer's opinion, sarcasm, dialect, rhetorical intent, and emojis. Then determine the stance toward the target itself, not toward other people, quoted opinions, related entities, or hashtags.

Guidelines

- Favor: supports, defends, promotes, or welcomes the target.
- Against: opposes, criticizes, rejects, or mocks the target.
- None: no clear stance toward the target.

Important

- Determine where praise or criticism is directed. Negative language toward opponents of the target is usually Favor, not Against.
- Hashtags may be ironic or hijacked. Never infer stance from hashtags alone.
- Rhetorical questions, sarcasm, and emojis often convey the writer's true stance. Interpret the intended meaning rather than the literal wording.
- Distinguish reporting from endorsement. Mentioning an event or policy does not by itself express a stance, unless it is framed positively (e.g. promoting, celebrating, or inviting participation), in which case it leans Favor.
- If the stance toward the target cannot reasonably be inferred, output None.

Respond with ONLY one word: Favor, Against, or None. Do not provide any explanation, punctuation, or additional text.

System Prompt - ALLaM-7B (Dialect Interpreter), Stage 1: Rephraser

You are an expert in Saudi Arabic dialect, sarcasm, and cultural context.

Given a tweet, explain what the writer literally means beneath the surface - resolve any sarcasm, dialect-specific phrasing, or rhetorical devices. Preserve the writer's emotional tone and attitude (e.g. celebratory, mocking, frustrated, joking, worried) - do not flatten it into a neutral factual summary. Focus on tone and intent, not on classifying any stance. Do not mention "Favor," "Against," or "None," and do not use evaluative words that imply approval or disapproval (e.g. "positive," "negative," "disapproving," "supportive") - describe the tone itself, not a judgment of it.

Stage 2 reuses the champion CI v3 prompt (Fig. A.2), consuming the Stage 1 interpretation in place of the raw tweet.

System Prompt - Explain-then-Classify, Stage 1: Explanation

You are an expert Arabic language analyst.

Background Context

The target concerns the 2017-2018 Saudi policy change allowing women to drive. Before June 2018, women were prohibited from driving in Saudi Arabia. Tweets from this period often discuss the royal decree, implementation, licensing, religion, tradition, safety, gender roles, media coverage, and public reaction. They frequently use Saudi dialect, sarcasm, humor, rhetorical questions, and indirect expressions.

Task

Given an Arabic tweet, rewrite its intended literal meaning in clear English, preserving the writer's opinion, sarcasm, dialect, rhetorical intent, and emojis. Do not classify stance. Do not mention Favor, Against, or None. Only explain what the tweet is actually saying and who or what it is directed at. Respond with plain explanation only.

System Prompt - Explain-then-Classify, Stage 2: Classification

You are an expert annotator for Arabic stance detection.

Background Context

The target concerns the 2017-2018 Saudi policy change allowing women to drive. Before June 2018, women were prohibited from driving in Saudi Arabia. Tweets from this period often discuss the royal decree, implementation, licensing, religion, tradition, safety, gender roles, media coverage, and public reaction. They frequently use Saudi dialect, sarcasm, humor, rhetorical questions, and indirect expressions.

Task

You will be given the original Arabic tweet and an explanation of its intended meaning. Classify the writer's stance toward the target as exactly one of:

- Favor
- Against
- None

Use the explanation to inform your reading, but weigh the original tweet's actual wording, tone, and emojis as well - the explanation is an aid, not a replacement for the tweet itself. Then determine the stance toward the target itself, not toward other people, quoted opinions, related entities, or hashtags.

Guidelines

- Favor: supports, defends, promotes, or welcomes the target.
- Against: opposes, criticizes, rejects, or mocks the target.
- None: no clear stance toward the target.

Important

- Determine where praise or criticism is directed. Negative language toward opponents of the target is usually Favor, not Against.
- Hashtags may be ironic or hijacked. Never infer stance from hashtags alone.
- Rhetorical questions, sarcasm, and emojis often convey the writer's true stance. Interpret the intended meaning rather than the literal wording.
- Distinguish reporting from endorsement. Mentioning an event or policy does not by itself express a stance, unless it is framed positively (e.g. promoting, celebrating, or inviting participation), in which case it leans Favor.
- If the stance toward the target cannot reasonably be inferred, output None.

Respond with ONLY one word: Favor, Against, or None. Do not provide any explanation, punctuation, or additional text.

System Prompt - GQG-Debate, Stage 1: Gemma Claim Generation

You are an expert annotator for Arabic stance detection.

Background Context

The target concerns the 2017–2018 Saudi policy change allowing women to drive. Before June 2018, women were prohibited from driving in Saudi Arabia. Tweets from this period often discuss the royal decree, implementation, licensing, religion, tradition, safety, gender

roles, media coverage, and public reaction. They frequently use Saudi dialect, sarcasm, humor, rhetorical questions, and indirect expressions.

Task

Given an Arabic tweet and a target topic, classify the writer's stance toward the target as exactly one of:

- Favor
- Against
- None

Before assigning a stance, mentally rewrite the tweet into its intended literal meaning while preserving the writer's opinion, sarcasm, dialect, rhetorical intent, and emojis. Then determine the stance toward the target itself, not toward other people, quoted opinions, related entities, or hashtags.

Guidelines

- Favor: supports, defends, promotes, or welcomes the target.
- Against: opposes, criticizes, rejects, or mocks the target.
- None: no clear stance toward the target.

Important

- Determine where praise or criticism is directed. Negative language toward opponents of the target is usually Favor, not Against.
- Hashtags may be ironic or hijacked. Never infer stance from hashtags alone.
- Rhetorical questions, sarcasm, and emojis often convey the writer's true stance. Interpret the intended meaning rather than the literal wording.
- Distinguish reporting from endorsement. Mentioning an event or policy does not by itself express a stance, unless it is framed positively (e.g. promoting, celebrating, or inviting participation), in which case it leans Favor.
- If the stance toward the target cannot reasonably be inferred, output None.

Respond in exactly this format:

Stance: <Favor/Against/None>

Argument: <justification for this stance>

System Prompt - GQG Debate, Stage 2: Qwen Challenge Generation

You are a critical reviewer of Arabic stance detection judgments.

You will be given a tweet, a target topic, and another annotator's classification with their justification. Your job is to challenge it:

- If you believe the classification is wrong, argue for what you believe the correct stance is (Favor, Against, or None) and explain why, citing specific words, tone, sarcasm, or context in the tweet.
- If you believe the classification is correct, still identify any weaknesses or alternative readings in the original justification - do not simply agree without scrutiny.

Be concise.

System Prompt - GQG-Debate, Stage 3: Gemma Judge Decision

You are an expert annotator for Arabic stance detection, making a final decision after reviewing two perspectives.

Background Context

The target concerns the 2017–2018 Saudi policy change allowing women to drive. Before June 2018, women were prohibited from driving in Saudi Arabia. Tweets from this period often discuss the royal decree, implementation, licensing, religion, tradition, safety, gender roles, media coverage, and public reaction. They frequently use Saudi dialect, sarcasm, humor, rhetorical questions, and indirect expressions.

You will be given a tweet, a target, an initial classification with its argument, and a challenge/critique of that classification. Weigh both perspectives on their merits - do not automatically favor the initial classification just because it came first.

Guidelines

- Favor: supports, defends, promotes, or welcomes the target.
- Against: opposes, criticizes, rejects, or mocks the target.
- None: no clear stance toward the target.

Important

- Determine where praise or criticism is directed. Negative language toward opponents of the target is usually Favor, not Against.
- Hashtags may be ironic or hijacked. Never infer stance from hashtags alone.
- Rhetorical questions, sarcasm, and emojis often convey the writer's true stance. Interpret the intended meaning rather than the literal wording.
- Distinguish reporting from endorsement. Mentioning an event or policy does not by itself express a stance, unless it is framed positively (e.g. promoting, celebrating, or inviting participation), in which case it leans Favor.
- If the stance toward the target cannot reasonably be inferred, output None.

Respond with ONLY one word: Favor, Against, or None. Do not provide any explanation, punctuation, or additional text.

A.3 Arabic and Stance Interpretation Guidelines

Table 4 summarizes the decision principles incorporated into the final prompting strategy, derived from development-set analysis. These were provided as reasoning constraints guiding the LLM's interpretation, not independent classifiers.

Guideline	Decision principle
Target resolution	Determine whether stance is directed toward the specified target rather than a related entity, event, or policy.
Sentiment $\neq$ stance	Negative or positive sentiment alone does not determine the stance label.
Reporting $\neq$ endorsement	Describing a policy, event, or announcement does not imply support for it.
Hashtag interpretation	Interpret hashtags jointly with the tweet rather than as direct stance labels.
Sarcasm	Consider whether literal wording conflicts with the author's intended position.
Rhetorical questions	Do not assume the literal interpretation of rhetorical questions.
Quoted opinions	Distinguish the author's position from opinions attributed to others.
Against calibration	Assign <i>Against</i> only when opposition is directed at the target.
Favor calibration	Do not infer <i>Favor</i> from factual or superficially positive language alone.
None	Prefer <i>None</i> when text does not establish a sufficiently clear stance.

Table 4: Arabic and stance interpretation guidelines.

A.4 Error Category Examples

Table 4 summarizes the calibrated decision principles; Examples 1–8 below illustrate the four recurring error categories discussed in Section 5 (§3.4) with representative development-set and test-set tweets (translated from Arabic).

Hashtag Hijacking.

Dev-Set Example (Row 350, ID 2354) - Target: COVID Vaccine - Gold: Favor

Translation: “The Sultanate delayed the vaccine, so you started a hashtag and said we don’t want a lockdown and we want a vaccine just like all other countries. The vaccine arrived and vaccination became mandatory for everyone, so you said we don’t want vaccination and started the hashtag #No_to_Mandatory_Vaccination. What do you want, oh great people?”

Why illustrative: The tweet hijacks the anti-vaccine protest hashtag to mock the public’s contradictory behavior. The actual stance is Favor (supporting the vaccine program).

Test-Set Example (Row 155) - Target: Women Driving | Illustrative Test-Set Case

Translation: “#TheKing_wins_for_leadership_of_women I felt relieved when you put on the turn signal hahaha... My sister driver: adjusting your niqab [face veil] will cause your car to roll over...”
Why illustrative: The tweet adopts the official campaign hashtag celebrating the royal decree but uses it to host a joke mocking women’s driving competence.

Sarcasm and Implicit Stance.

Dev-Set Example (Row 570, ID 3806) - Target: Women Empowerment - Gold: Favor

Translation: “I feel this guy is one of those traditionalists (Sahwa members) whose trend has expired; he put the flag on his profile and pretends to be a patriot every now and then, while in reality, he opened his account to cry about women’s empowerment and wants women to go back to being buried like before the MBS era. Oh well, I hope I am not misjudging him and everything we see is just ‘lapses’ ”
Why illustrative: Rather than stating support directly, the writer uses sarcastic insults to mock a traditionalist opponent who hypocritically claims patriotism. The underlying stance is implicitly Favor.

Test-Set Example (Row 312) - Target: Women Driving | Illustrative Test-Set Case

Translation: “#You_will_not_drive Who is this still holding out writing ‘you won’t drive’? Mashallah, may God strengthen you. Keep going, my love ”
Why illustrative: The literal text appears to encourage anti-driving campaigners, but the heavy use of laughing emojis and skeptical phrasing signal sarcasm.

Reporting versus Endorsement.

Dev-Set Example (Row 447, ID 3057) - Target: Digital Transformation - Gold: None

Translation: “Al-Shawabkeh... Opening 4 projects for the Rapid Transit Bus and a 93% completion rate for the digital transformation project [URL]”
Why illustrative: A factual news announcement concerning infrastructure milestones, with no subjective stance expressed.

Test-Set Example (Row 19) - Target: Women Driving | Illustrative Test-Set Case

Translation: “Security forces distribute roses to ladies during the beginning of their driving cars in Saudi Arabia. . #Saudi_women_drive #SaudiWomenDriving”
Why illustrative: Describes a factual event (police distributing flowers to women driving on the first day of the policy) without subjective praise or opposition - classified as neutral news description (None).

Historical and Cultural Context.

Dev-Set Example (Row 2, ID 2) - Target: Women Empowerment - Gold: Against

Translation: “The situation in Saudi Arabia has become embarrassing-no sane companies left, nor sane celebrities. The current situation is the ugliest form of commodifying women under the guise of ‘women’s empowerment’! #Boycott_Khedeir_Tea”
Why illustrative: References the 2022 marketing controversy surrounding “Khedeir Tea,” criticized for exploitative use of women in ads. The writer attacks corporate initiatives, labeling them superficial commodification disguised as empowerment - stance is Against.

Test-Set Example (Row 0) - Target: Women Driving | Illustrative Test-Set Case

Translation: “We must send you a veiled Kuwaiti woman from our side whose car is a Corolla, to serve as a trial sample. And if you are convinced, I will shave off my mustache with a piece of glass #Saudi_Woman_Driving”
Why illustrative: Reflects pre-2018 public skepticism prior to the policy’s execution. The writer proposes borrowing a Kuwaiti woman (Kuwait already permitted female driving) as a test case, using a traditional Arab hyperbolic vow of disbelief. Without regional-driving-law context and cultural knowledge of such vows, the tweet is uninterpretable.

A.5 Fine-Tuning Configuration

Table 5 summarizes AraBERT training configuration. Class-weighted loss reduced the effect of label imbalance in the training data; checkpoints were selected on the held-out stratified validation split.

Parameter	Configuration
Models	AraBERTv0.2-Base, Twitter-Base, Twitter-Large
Input format	[CLS] Target [SEP] Tweet [SEP]
Preprocessing	AraBERTPreprocessor
Loss	Class-weighted cross-entropy
Optimizer	AdamW
Learning rate	2e-5
Batch size	16
Max epochs	5
Early stopping	Patience 2
Validation	90/10 stratified split
Small models (HW)	RTX 3050, 4GB

Table 5: AraBERT fine-tuning configuration.